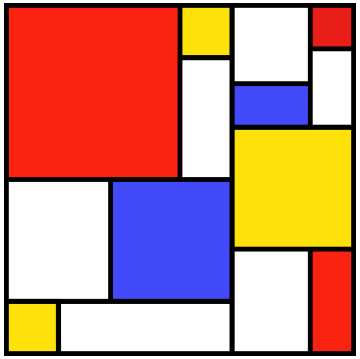
# SLiM

## Workshop Series



## Demography

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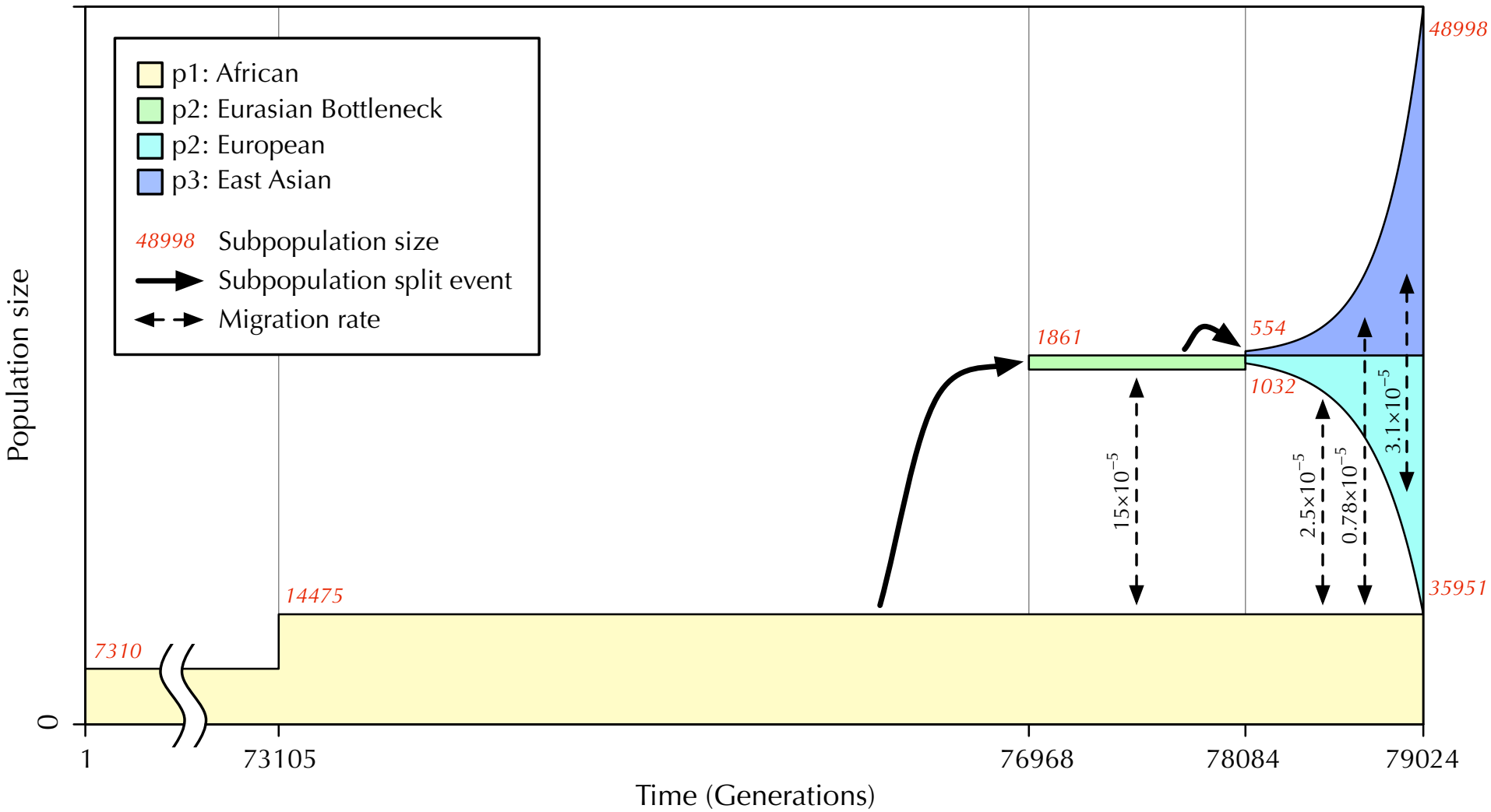
# Demography

- Demographic events in SLiM:
  - new subpopulation: `addSubpop()`
  - split a subpopulation: `addSubpopSplit()`
  - change in size: `setSubpopulationSize()`
  - remove a subpop: `setSubpopulationSize(0)`
  - set migration rate: `setMigrationRates()`
  - merge subpops:

    `addSubpop() + setMigrationRates()`

# The Gravel Model



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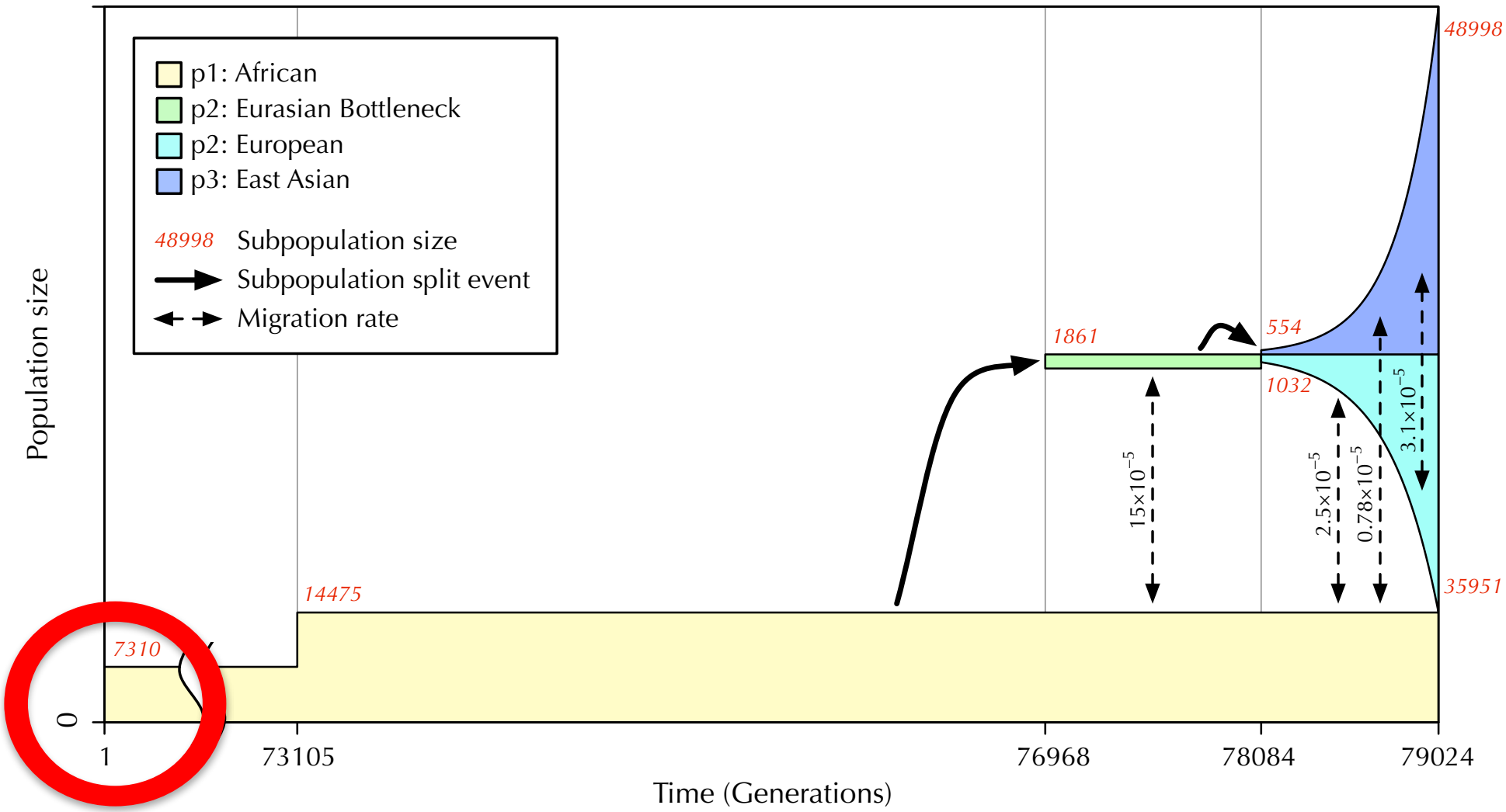

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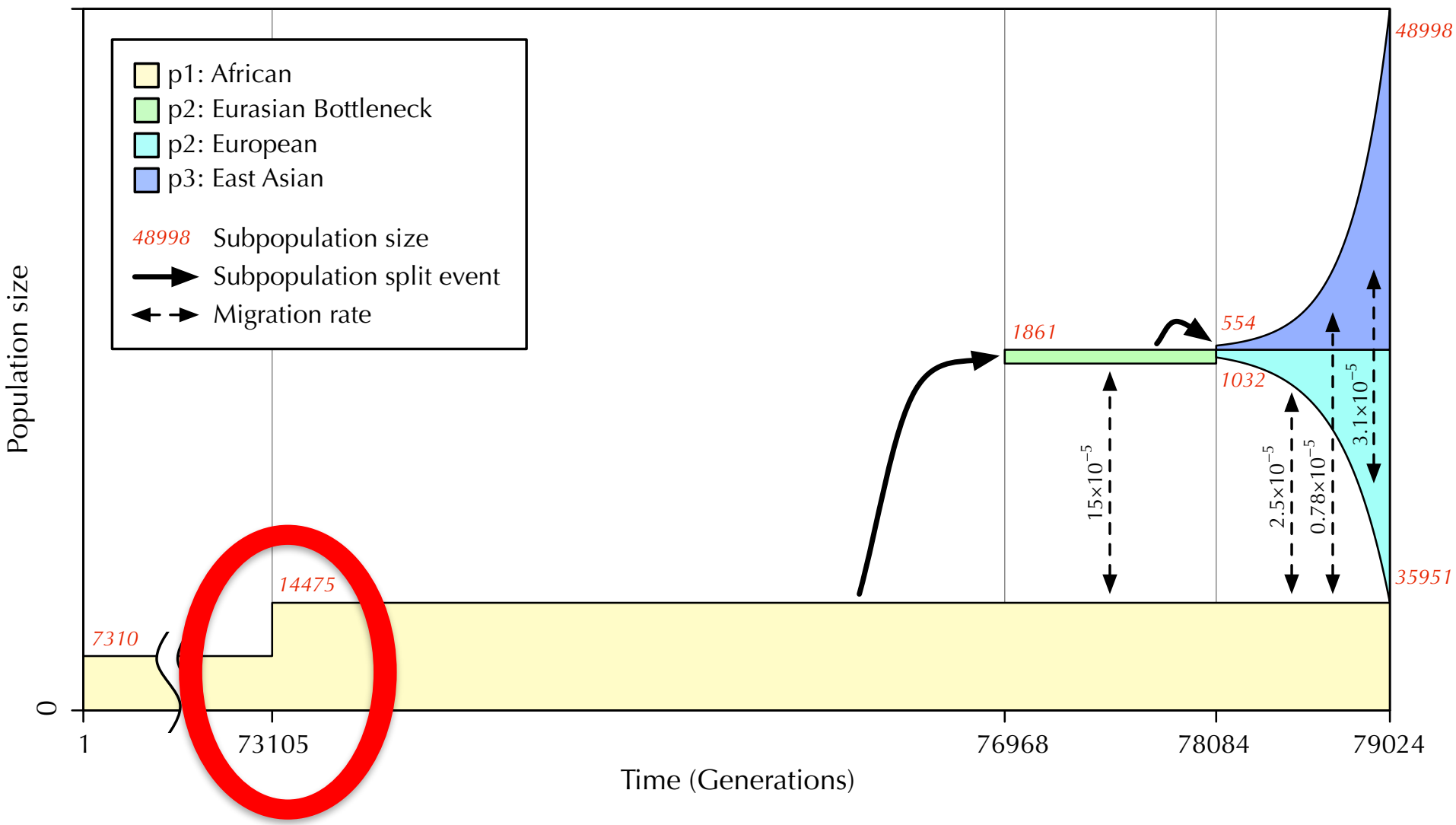

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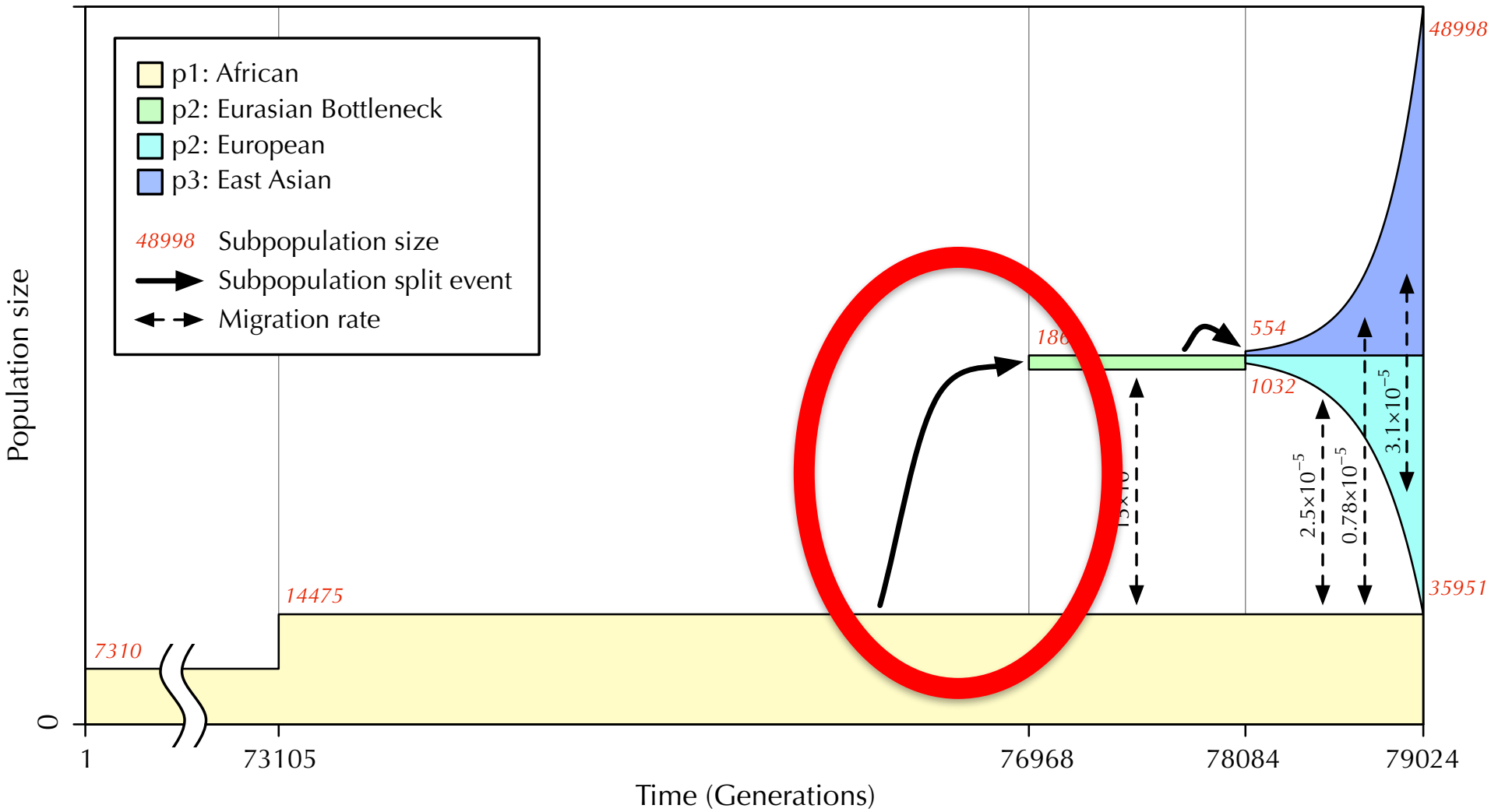

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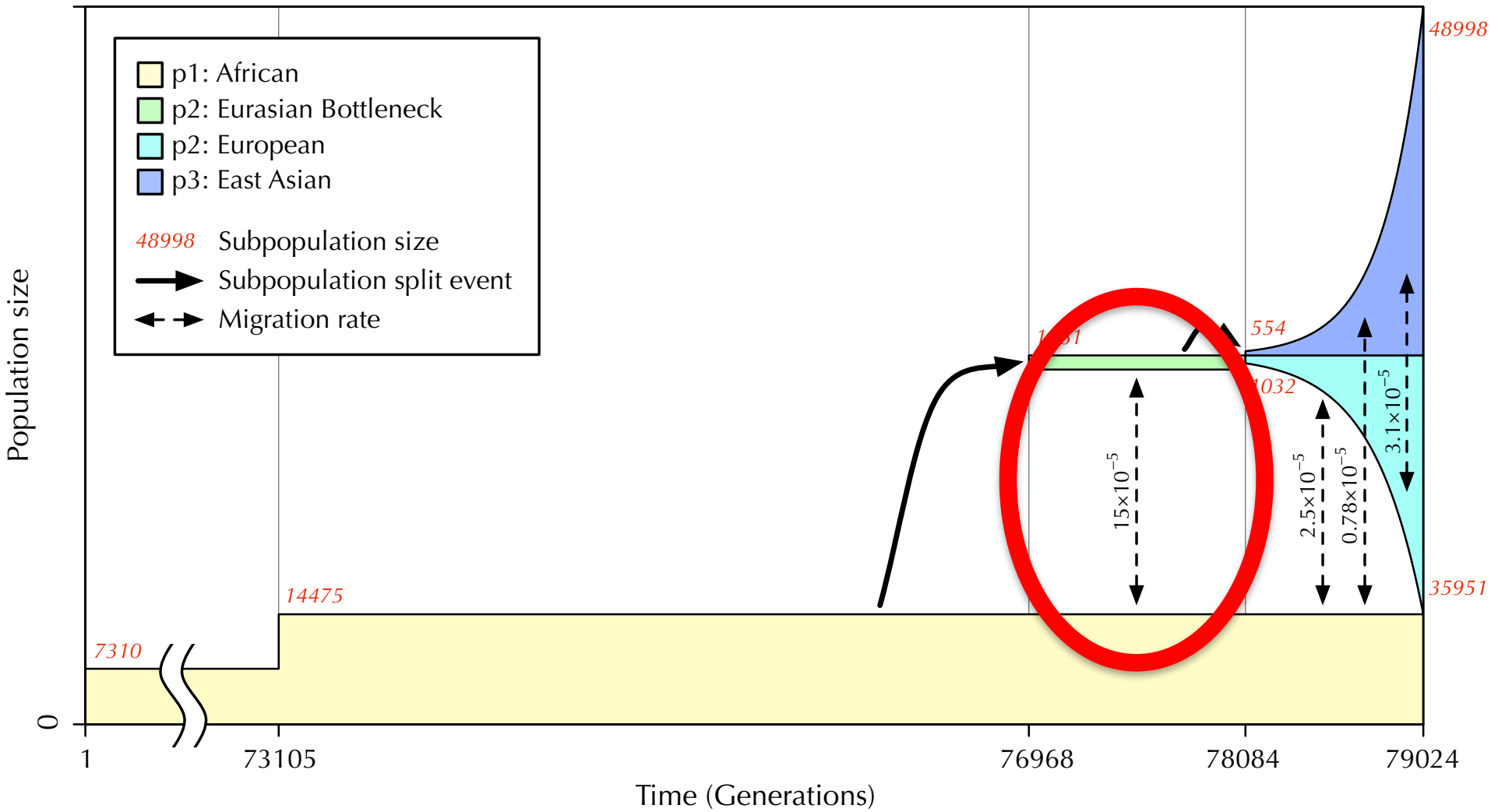

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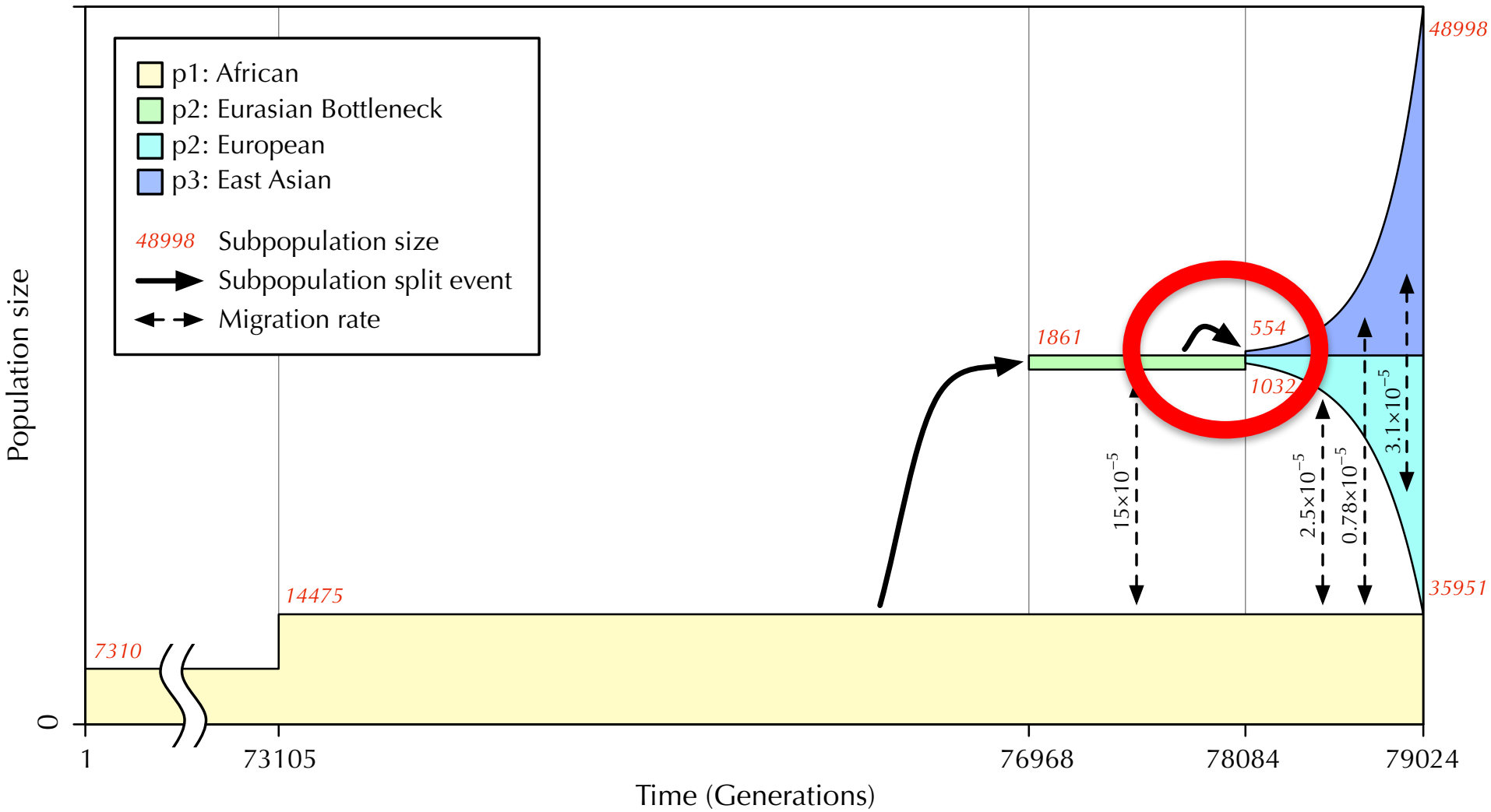

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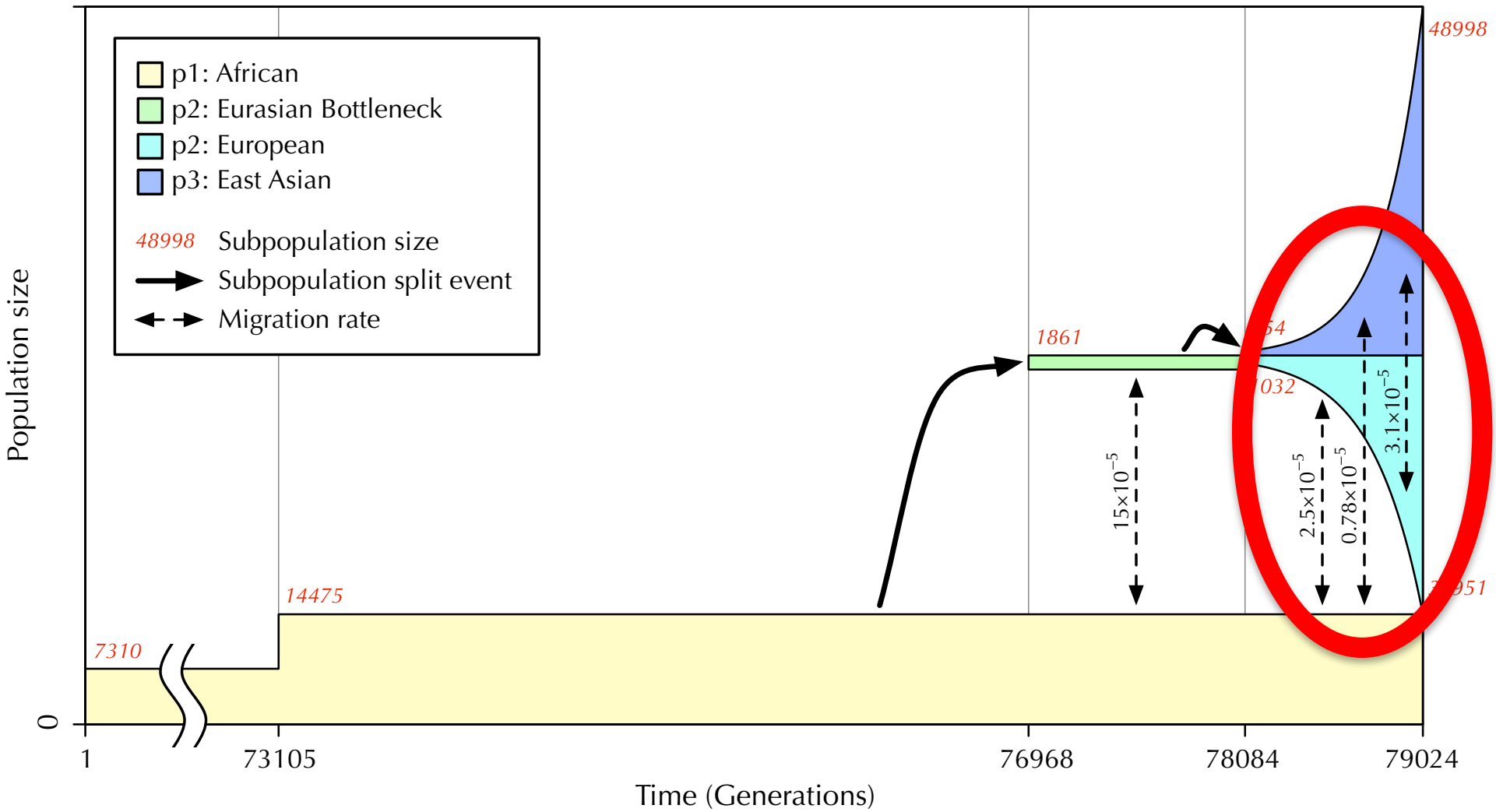

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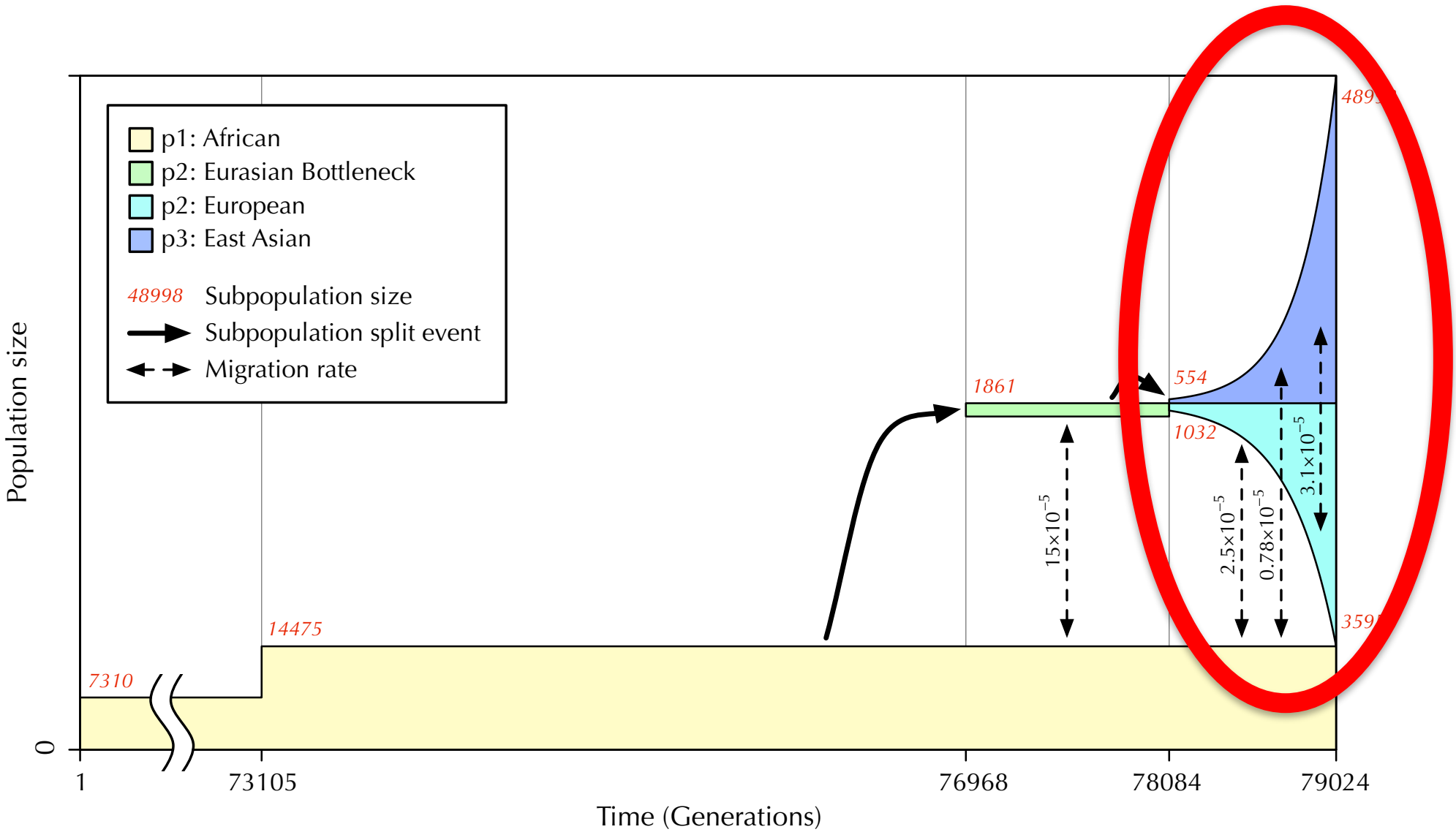

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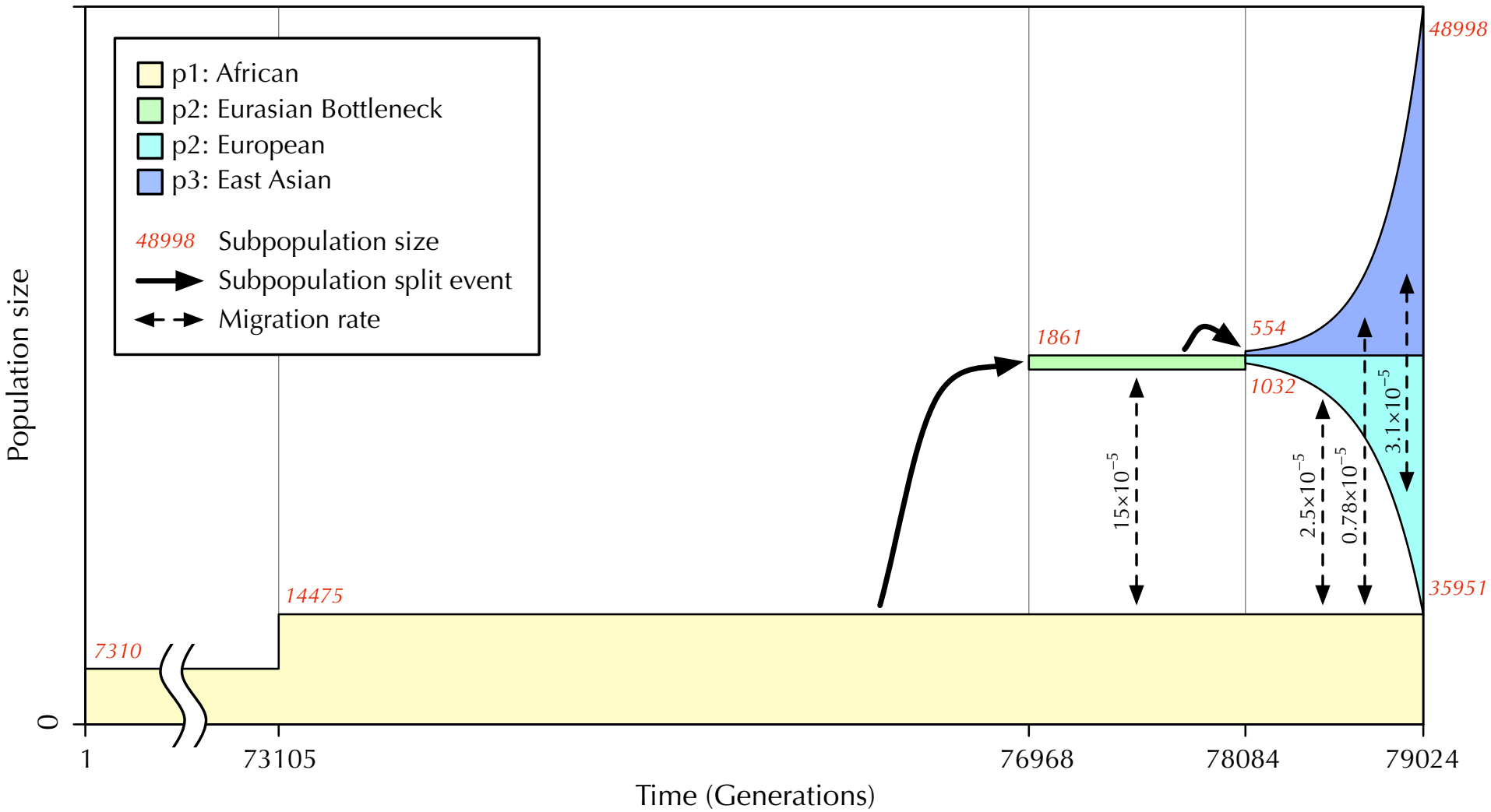

# Mating Systems

- Demographic parameters in SLiM:
  - enabling separate sexes: `initializeSex()`
  - sex ratio: `setSexRatio()`
  - cloning rate: `setCloningRate()`
  - selfing rate: `setSelfingRate()`

# Mating Systems

- Demographic parameters in SLiM:
  - enabling separate sexes: `initializeSex()`
  - sex ratio: `setSexRatio()`
  - cloning rate: `setCloningRate()`
  - selfing rate: `setSelfingRate()`
- nonWF models are more flexible:
  - monogamous mating
  - alternation of generations
  - haplodiploidy

*The WF model's tick cycle*

0. Execution of `first()` events

1. Execution of `early()` events

2. Generation of offspring:

   2.1. Choose source subpop

   2.2. Choose parent 1

   2.3. Choose parent 2 (`mateChoice()` callbacks)

   2.4. Generate the offspring (including `mutation()` and `recombination()` callbacks)

   2.5. Suppress/modify child (`modifyChild()` callbacks)

3. Removal of fixed mutations

4. Offspring become parents

5. Execution of `late()` events

6. Fitness recalculation using `mutationEffect()` and `fitnessEffect()` callbacks

7. Tick/cycle count increment

# The WF Tick Cycle

- The **tick cycle**
  - repeated by SLiM each tick
  - `first()` events (uncommon)
  - `early()` events
  - offspring generation
  - `late()` events
  - fitness recalculation

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- The **tick cycle**
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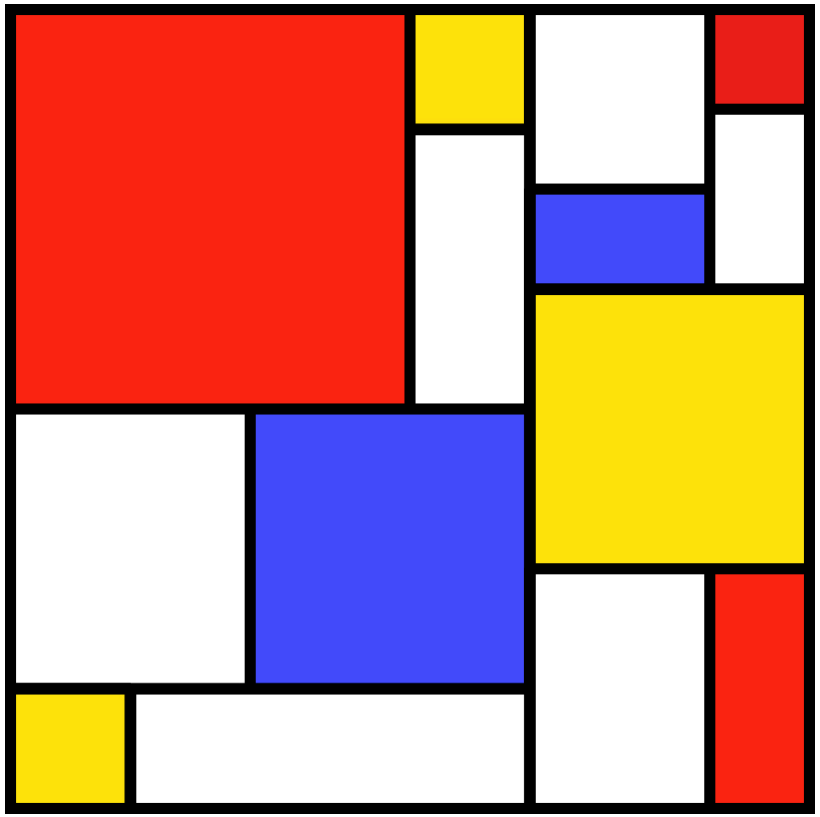
  - `first()` events (uncommon)
  - `early()` events
  - offspring generation
  - `late()` events
  - fitness recalculation
- **Future requests**
  - many changes take effect the next time offspring are generated!



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